BioGen NL Agent – Phase 1 Report

Data Parsing, Indexing and Search

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Natural Language Processing and Search
2025/2026

April 13, 2026

1 Introduction

Large language models have significantly improved natural language understanding and generation in the biomedical domain. However, in high-stakes settings such as health-related question answering, factual grounding remains essential. A retrieval component is therefore a fundamental part of a reliable biomedical natural language agent, as it determines which scientific evidence is available to later stages of the system.

In this first phase, we focus on the retrieval layer of the BioGen NL Agent. Given a biomedical topic, the system must search a collection of PubMed articles and return the most relevant documents. We study different retrieval strategies and compare their effectiveness using standard information retrieval metrics. This phase establishes the experimental foundation for the following stages.

2 BioGen

2.1 Data Parsing, Indexing and Search

2.1.1 Corpus and Topic Data

The document collection consists of PubMed records provided in a text file, where each line is a JSON object containing a document identifier and a field named `contents`. The first line of `contents` corresponds to the article title and the remaining text corresponds to the abstract. During preprocessing, each record was parsed into the following fields:

- `pmid`: PubMed identifier;
- `title`: article title;
- `abstract`: article abstract;

The topic file contains biomedical information needs represented by three fields:

- `topic`;
- `question`;
- `narrative`.

For retrieval, the final query text was formed by concatenating these three fields into a single query string.

2.1.2 Relevance Judgments

Relevance judgments were derived from the provided submission annotations. For each question, cited PMIDs in accurate system answers were inspected. Citations labeled as *supporting* were assigned relevance score 2, citations labeled as *neutral* were assigned relevance score 1, and citations labeled as *contradicting* or *not relevant* assigned relevance score 0. This produced graded relevance judgments suitable for NDCG evaluation, as well as binary relevance sets for precision, recall, MAP, and MRR.

2.1.3 Indexing in OpenSearch

The retrieval system was implemented in OpenSearch. The index stores lexical fields with different similarity functions as well as a dense vector field for KNN search. The main indexed fields are:

- `pmid`;
- `title`;
- `abstract`;
- `bm25_content`;
- `jelinek_content`;
- `dirichlet_content`;
- `embedding`.

The three lexical content fields contain the same concatenated title and abstract text, but each field uses a different retrieval model. This allows a fair comparison of lexical ranking functions over the same document representation.

2.1.4 Retrieval Models

We implemented four retrieval strategies:

1. **BM25**: standard lexical ranking baseline.
2. **Language Model with Jelinek-Mercer smoothing (JM)**.

3. **Language Model with Dirichlet smoothing (LMD)**.
4. **KNN dense retrieval**: nearest-neighbor retrieval over document embeddings.

For KNN retrieval, document and query texts were encoded into dense vectors using a sentence embedding model, and OpenSearch was used to retrieve the nearest documents in vector space.

2.1.5 Implementation Pipeline

The Phase 1 pipeline is summarized as follows:

1. Parse PubMed records into structured fields.
2. Build an OpenSearch index with lexical and dense representations.
3. Encode documents for vector search.
4. Retrieve the top-k documents for each topic using each retrieval model.
5. Evaluate the ranked lists against the relevance judgments.

3 Evaluation

3.1 Experimental Setup: Datasets, Metrics and Protocols

3.1.1 Datasets

The experiments use:

- a PubMed abstract corpus with 4194 indexed documents;
- the BioGen topic set;
- graded relevance judgments derived from the provided submission annotations.

3.1.2 Metrics

The retrieval models were evaluated using the following metrics:

- **Precision@10 (P@10)**: proportion of relevant documents among the top 10 retrieved results;
- **Recall@100 (R@100)**: proportion of all relevant documents retrieved in the top 100 results;
- **NDCG@10 and NDCG@100**: ranking quality under graded relevance;
- **MAP**: overall ranking effectiveness across all relevant documents;
- **MRR**: inverse rank of the first relevant document.

In addition, precision-recall curves were generated to compare ranking behavior across cutoffs.

3.1.3 Protocol

Each model was run over the topic set and evaluated first over all topics for exploratory analysis and then over the test split for final comparison. The test split results are the main basis for the final conclusions in this phase.

3.2 Results and Discussion

3.2.1 Main Quantitative Results

Table 1 presents the main retrieval results on the test set.

Table 1: Phase 1 test set retrieval results.

Method	P@10	R@100	NDCG@10	NDCG@100	MAP
BM25	0.6667	0.8435	0.6417	0.7652	0.5979
JM	0.6758	0.8080	0.6503	0.7404	0.5693
LMD	0.6636	0.8104	0.6315	0.7374	0.5659
KNN	**0.7485**	**0.8946**	**0.7207**	**0.8247**	**0.6719**

3.2.2 Discussion

The results show a clear performance gap between dense retrieval and lexical baselines. KNN achieved the best results in all main evaluation metrics on the test set, including Precision@10, Recall@100, NDCG@10, NDCG@100, and MAP. This suggests that dense retrieval is better able to capture semantic similarity between biomedical questions and article abstracts, even when exact lexical overlap is limited.

Among the lexical methods, BM25 was the strongest baseline. It outperformed both JM and LMD on most of the reported test metrics. In particular, BM25 achieved a strong Recall@100 and competitive NDCG values, indicating that simple term-based matching remains effective in biomedical retrieval when topic wording overlaps with the document content.

Jelinek-Mercer and Dirichlet smoothing produced weaker overall results in this setup. Their performance remained reasonable, but they did not surpass BM25 on the main test metrics. This may indicate that the corpus, query formulation, and OpenSearch deafault configuration favor BM25's ranking behavior for this task.In the following section, we will elaborate on how this values can be improved by tweaking the hyperparameters to our specific case.

An interesting detail is that BM25 achieved the highest MRR on the test set, meaning that it often placed at least one relevant document very early in the ranking. However, KNN retrieved more relevant documents overall and produced stronger ranking quality across the top positions, which makes it the best candidate retrieval method for the next phase.

The mean Average Precision (mAP) across all test queries was 0.6719 for the KNN model, which was the highest among the evaluated methods.

3.2.3 Precision-Recall Analysis

Figure 1 includes the precision-recall curves of the four retrieval methods.

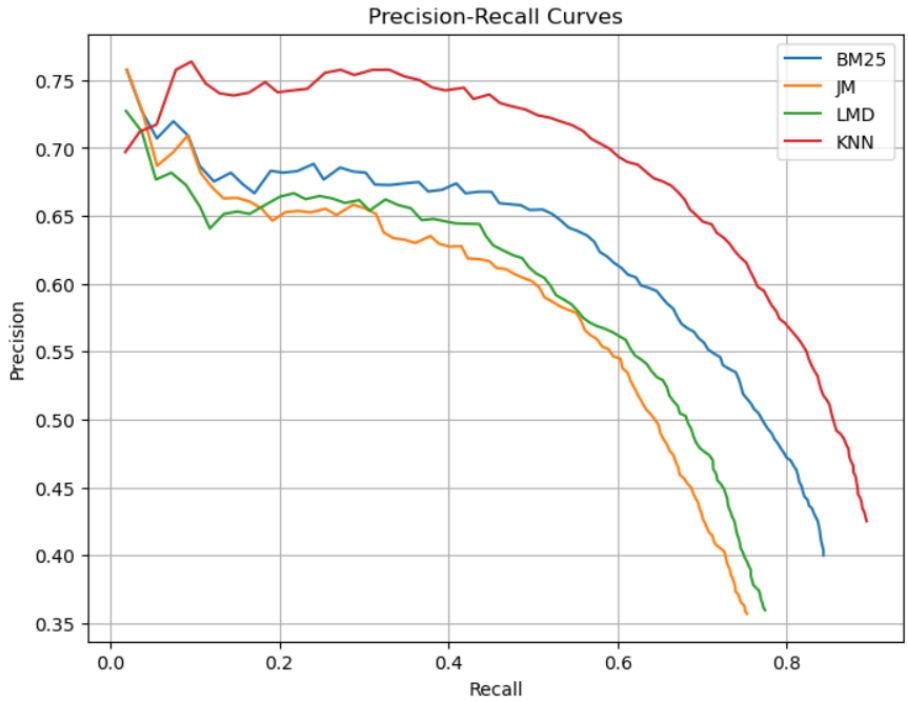

Figure 1: Precision-recall curves for the evaluated retrieval methods.

The precision-recall curves are expected to reflect the same overall tendency observed in the table, with KNN maintaining higher precision across a wider recall range than the lexical baselines.

3.2.4 Query-Level Precision-Recall Analysis

To further analyze system stability, we examine three representative queries: the query with the highest Average Precision (AP), the query with the lowest AP, and a query with median performance.

Figure 2 presents the precision-recall curves for these three queries using the KNN retrieval model.

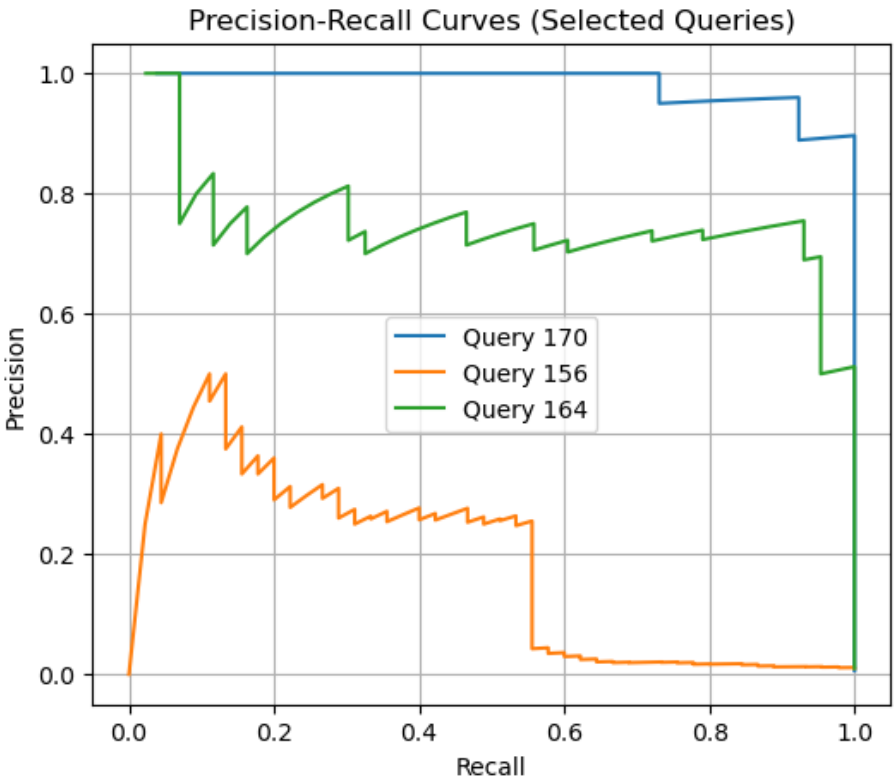

Figure 2: Precision-recall curves for selected queries (best, worst, and median AP).

The query with the highest AP (Topic **170**) maintains near-perfect precision for a large portion of the recall range, with precision remaining close to 1.0 before slightly decreasing at higher recall levels. This indicates that relevant documents are retrieved very early and consistently ranked at the top positions, with only a small number of non-relevant documents appearing later in the ranking.

In contrast, the query with the lowest AP (Topic **156**) exhibits low precision across almost the entire recall range. Although there is a small increase in precision at very low recall levels, it quickly drops and remains low, approaching zero as recall increases. This suggests that relevant documents are either scarce, difficult to retrieve, or ranked poorly, leading to weak overall performance.

The intermediate query (Topic **164**) shows a more balanced behavior. Precision starts high but decreases gradually as recall increases, with noticeable fluctuations throughout the curve. This indicates that relevant documents are retrieved early, but non-relevant documents are increasingly mixed into the ranking as more results are considered.

These results highlight that retrieval performance varies significantly across queries. While some queries are handled very effectively, others remain challenging due to factors such as vocabulary mismatch or semantic complexity, reinforcing the importance of robust retrieval methods for biomedical information needs.

3.3 Hyperparameter Analysis

The default OpenSearch smoothing parameters ($\lambda = 0.1$, $\mu = 2000$) apply very heavy corpus smoothing. Given our small corpus of 4,194 documents with an average abstract length of 139 words, this is poorly suited — the corpus background is not reliable enough to dominate scoring. We therefore evaluated two alternative configurations on the training split: $\lambda = 0.2/\mu = 150$ and $\lambda = 0.5/\mu = 150$, setting μ near the average document length to balance document and corpus contributions.

Tables 2, 3, and 4 show the results for each configuration.

Table 2: Default settings ($\lambda = 0.1$, $\mu = 2000$).

Method	MAP	MRR	P@10	R@100	NDCG@10	NDCG@100
KNN	0.6719	0.8258	0.7485	0.8946	0.7207	0.8247
BM25	0.5979	0.8636	0.6667	0.8435	0.6417	0.7652
LMD	0.5353	0.8535	0.6515	0.7745	0.6144	0.7076
JM	0.5263	0.8503	0.6606	0.7531	0.6309	0.6976

Table 3: Tuned settings ($\lambda = 0.2$, $\mu = 150$).

Method	MAP	MRR	P@10	R@100	NDCG@10	NDCG@100
KNN	0.6719	0.8258	0.7485	0.8946	0.7207	0.8247
BM25	0.5979	0.8636	0.6667	0.8435	0.6417	0.7652
LMD	0.5659	0.8662	0.6636	0.8104	0.6315	0.7374
JM	0.5420	0.8561	0.6545	0.7700	0.6296	0.7117

Table 4: Tuned settings ($\lambda = 0.5$, $\mu = 150$).

Method	MAP	MRR	P@10	R@100	NDCG@10	NDCG@100
KNN	0.6719	0.8258	0.7485	0.8946	0.7207	0.8247
BM25	0.5979	0.8636	0.6667	0.8435	0.6417	0.7652
JM	0.5693	0.8495	0.6758	0.8080	0.6503	0.7404
LMD	0.5659	0.8662	0.6636	0.8104	0.6315	0.7374

Both JM and LMD improve consistently as parameters move away from the defaults. The best configuration, $\lambda = 0.5/\mu = 150$, yields NDCG@10 gains of +3.1% for JM and +2.8% for LMD over the defaults.

4 Conclusion

4.1 Achievements

In Phase 1, we developed the retrieval component of the BioGen NL Agent. We parsed and indexed a PubMed-based corpus in OpenSearch, implemented four retrieval strategies, and evaluated them using standard information retrieval metrics. The experiments showed that dense retrieval with KNN embeddings delivered the strongest overall performance, while BM25 was the best lexical baseline, even after hyperparameter tuning.

This phase establishes a solid retrieval layer for the next stage of the project, where the retrieved biomedical documents will be used to support answer generation in a retrieval-augmented generation setting.

4.2 Limitations

This phase has several limitations. First, relevance judgments were derived from submission annotations rather than from a dedicated manually curated retrieval qrels file, which may introduce inconsistencies into the evaluation. Second, the experiments used a single query formulation based on the concatenation of topic, question, and narrative, so other formulations may achieve better outcome and improve retrieval effectiveness.

Future work in the next phase will build on the best retrieval configuration and investigate how to select supporting evidence spans and generate grounded biomedical answers.